SUBCHAPTER 3.2 · CONTINUITY BALANCES

Conservation of Mass (Continuity) in Problem Solving

Global mass balances, 1D streamtube formulations, unsteady tank drainage, and upstream streamtube capture width determination.

1. The Master Integral Continuity Equation

In classical fluid mechanics, matter is neither created nor destroyed. For any closed material fluid system $V_f(t)$, the rate of change of mass is identically zero:

$$\frac{dM_{\text{sys}}}{dt} = \frac{d}{dt}\left[\int_{V_f(t)} \rho \, dV\right] = 0$$

[Eq. 3.9] Axiom of mass conservation for a material fluid system.

Setting the intensive property $b = 1$ in the universal Reynolds Transport Theorem (Eq. 3.8) yields the **Master Integral Mass Conservation Equation** for any moving control volume $V_c(t)$:

$$\mathbf{\frac{d}{dt}\left[\int_{V_c(t)} \rho \, dV\right] + \int_{\Sigma_c(t)} \rho[(\vec{v} - \vec{v}_c)\cdot\vec{n}]\, d\sigma = 0}$$

[Eq. 3.10] Integral continuity equation for an arbitrary moving control volume.

For a fixed, rigid control volume ($V_0, \vec{v}_c = \vec{0}$), this simplifies to the Eulerian storage and convective flux balance:

$$\int_{V_0} \frac{\partial \rho}{\partial t}\, dV + \int_{\Sigma_0} \rho(\vec{v}\cdot\vec{n})\, d\sigma = 0$$

[Eq. 3.11] Continuity equation for a fixed spatial control volume.

Mass Conservation: Complete Derivation Chain (Notes.pdf, Eqs. 3.9-3.11 and 4.4-4.7)

Step 1: Mass balance for an arbitrary moving control volume (Eq. 3.10)

The principle (Eq. 3.9) states that the mass of a fluid volume does not change, $dM/dt = 0$ with $M = \int_{V_f(t)} \rho \, dV$. Choose $\phi = \rho$ in the Reynolds Transport Theorem (Eq. 3.8) and replace the left-hand side by zero:

$$0 = \frac{d}{dt} \int_{V_f(t)} \rho \, dV = \frac{d}{dt} \int_{V_c(t)} \rho \, dV + \int_{\Sigma_c(t)} \rho \, (\vec{v} - \vec{v}_c) \cdot \vec{n} \, d\sigma$$

Moving the flux term to the other side shows the physical meaning: the rate of change of the mass stored in the control volume equals minus the net mass flux out through its boundary, that is, the mass flux entering.

$$\frac{d}{dt} \int_{V_c(t)} \rho \, dV = - \int_{\Sigma_c(t)} \rho \, (\vec{v} - \vec{v}_c) \cdot \vec{n} \, d\sigma$$

Eq. 3.10, rearranged

Step 2: Fixed control volume (Eq. 3.11)

For a fixed volume V_0 with surface Σ_0 we have $\vec{v}_c = \vec{0}$ and the domain does not depend on t. The limits of integration are therefore constants, no boundary term appears, and the time derivative passes inside the integral (compare Eq. 3.7 with $\vec{v}_c = \vec{0}$):

$$\frac{d}{dt} \int_{V_0} \rho \, dV = \int_{V_0} \frac{\partial \rho}{\partial t} \, dV$$

Substituting into Eq. 3.10 with $\vec{v}_c = \vec{0}$ gives

$$\int_{V_0} \frac{\partial \rho}{\partial t} \, dV + \int_{\Sigma_0} \rho \, \vec{v} \cdot \vec{n} \, d\sigma = 0$$

Eq. 3.11

Step 3: Mass flow rate and sign convention

With $\vec{n}$ outward, the algebraic mass flow rate through a part A of the control surface is

$$\dot{m}_A = \int_A \rho\,(\vec{v} - \vec{v}_c)\cdot\vec{n}\,d\sigma$$

It is positive at outlets, negative at inlets and zero on impermeable walls, where no fluid crosses the surface and $(\vec{v} - \vec{v}_c)\cdot\vec{n} = 0$. Eq. 3.10 then reads $dM_c/dt = -\sum_A \dot{m}_A$. For a steady flow in a fixed volume the left-hand side vanishes and $\sum \dot{m}_{\text{out}} = \sum \dot{m}_{\text{in}}$ (magnitudes). On a port of area A_k with uniform ρ_k and uniform velocity v_k normal to the port, $|\dot{m}_k| = \rho_k v_k A_k$.

Step 4: Differential form (Eqs. 4.4-4.7 of Notes.pdf, Chapter 4)

The following steps belong to Chapter 4 of the Notes and are reproduced here to close the logical chain between the integral and the local statement. Apply Eq. 3.6 with $\phi = \rho$ to Eq. 3.9:

$$\int_{V_f(t)} \frac{\partial \rho}{\partial t}\,dV + \int_{\Sigma_f(t)} \rho\,\vec{v}\cdot\vec{n}\,d\sigma = 0$$

Gauss' theorem (Eq. 2.9), $\int_\Sigma \vec{A}\cdot\vec{n}\,d\sigma = \int_V \nabla\cdot\vec{A}\,dV$ with $\vec{A} = \rho\vec{v}$, turns the surface integral into a volume integral over the same domain:

$$\int_{V_f(t)} \left[\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\vec{v})\right] dV = 0$$

Eq. 4.4

Arbitrary-volume argument. Let $g(\vec{x},t) = \partial\rho/\partial t + \nabla\cdot(\rho\vec{v})$, a continuous function. Eq. 4.4 holds for every fluid volume. Suppose $g(\vec{x}^*,t) \neq 0$ at some point, say $g(\vec{x}^*,t) > 0$. By continuity $g > 0$ in a ball $B_\varepsilon(\vec{x}^*)$. The fluid volume that occupies this ball at time t then gives $\int_{B_\varepsilon} g\,dV > 0$, which contradicts Eq. 4.4. The case $g < 0$ is identical. Hence $g \equiv 0$ at every point:

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0$$

Eq. 4.5

Expand the divergence, $\nabla\cdot(\rho\vec{v}) = \vec{v}\cdot\nabla\rho + \rho\,\nabla\cdot\vec{v}$, and use the material derivative $D\rho/Dt = \partial\rho/\partial t + \vec{v}\cdot\nabla\rho$:

$$\frac{1}{\rho}\frac{D\rho}{Dt} = -\nabla \cdot \vec{v}$$

Eq. 4.6

For a constant-density fluid $D\rho/Dt = 0$, so $\nabla \cdot \vec{v} = 0$ (Eq. 4.7). For a steady flow of a gas, $\partial\rho/\partial t = 0$ and Eq. 4.5 gives $\nabla \cdot (\rho\vec{v}) = 0$ (Eq. 4.8).

Step 5: Control volume bounded by a moving free surface (extension)

Extension, not in the Notes. For a tank, the free surface $z = h(t)$ is part of Σ_c and moves with $\vec{v}_c = \dot{h}\,\vec{e}_z$. It is a material surface (no fluid crosses it), so the kinematic condition $\vec{v} \cdot \vec{n} = \vec{v}_c \cdot \vec{n}$ holds. Therefore $(\vec{v} - \vec{v}_c) \cdot \vec{n} = 0$ on it and it contributes no flux to Eq. 3.10. Only the inlet and the outlet contribute. With constant density and a tank of constant horizontal area A_T, the stored mass is $\rho A_T h(t)$ and Eq. 3.10 gives

$$\rho A_T \frac{dh}{dt} = \rho Q_{\text{in}} - \rho Q_{\text{out}} \implies A_T \frac{dh}{dt} = Q_{\text{in}} - Q_{\text{out}}$$

The discharge law $Q_{\text{out}} = A_e\sqrt{2gh}$ (Torricelli) is an input of the problem. It follows from Bernoulli's equation for steady inviscid flow, which is not derived in Topic 3 of the Notes, so it is taken here as an assumption.

2. Incompressible Fluids & 1D Streamtubes

When the fluid density is spatially and temporally constant ($\rho = \text{const}$, valid for liquids and for gas flows where the Mach number $Ma = v/a < 0.3$), the density factors out of both integrals. For a fixed control volume:

$$\int_{\Sigma_0} \vec{v} \cdot \vec{n}\, d\sigma = 0 \implies \sum_k Q_{\text{in},k} = \sum_j Q_{\text{out},j}$$

where $Q = \int_A (\vec{v} \cdot \vec{n}) dA$ is the volumetric flow rate (m^3/s).

1D Streamtube Approximation & Average Velocity

Across any duct or streamtube cross-section A, the mass flow rate $\dot{m}$ (kg/s) is:

$$\dot{m} = \int_A \rho(\vec{v} \cdot \vec{n})\, dA = \rho \bar{v} A$$

where the **area-averaged bulk velocity** $\bar{v}$ is defined by:

$$\bar{v} = \frac{1}{A}\int_A v(\vec{x})\, dA$$

> **VELOCITY PROFILES & BULK VELOCITY**
> In laminar circular pipe flow (Poiseuille), $v(r) = v_{\max}[1-(r/R)^2]$, so the bulk velocity is $\bar{v} = \frac{1}{2}v_{\max}$. In turbulent pipe flow, the profile is fuller ($1/7$th power law), yielding $\bar{v} \approx 0.817 v_{\max}$. Always integrate the profile explicitly when given!

3. Unsteady Storage: Filling & Emptying Tanks

When the mass inside the control volume changes with time, the unsteady storage term must be evaluated by tracking the moving liquid level $h(t)$.

Consider a tank of horizontal cross-sectional area A_T with an inlet flow rate Q_{in} and a bottom discharge orifice of area $A_e \ll A_T$. Defining a control volume that encloses the liquid inside the tank up to the free surface $z = h(t)$:

$$\frac{d}{dt}\left[\int_{V_c}\rho\, dV\right] = \rho\frac{d}{dt}\left[A_T h(t)\right] = \rho A_T\frac{dh}{dt}$$

Net mass flux across the boundaries:

$$\int_{\Sigma_c}\rho(\vec{v}\cdot\vec{n})\, d\sigma = \rho Q_{\text{out}} - \rho Q_{\text{in}}$$

Applying Torricelli's law for inviscid discharge from the bottom orifice, $v_e(t) = \sqrt{2gh(t)}$, so $Q_{\text{out}} = A_e\sqrt{2gh(t)}$. Substituting into the continuity equation:

$$A_T\frac{dh}{dt} = Q_{\text{in}} - A_e\sqrt{2gh(t)}$$

For a draining tank with no inflow ($Q_{\text{in}} = 0$) and initial height H_0, separating variables and integrating yields the classical draining time:

$$\int_{H_0}^{0} \frac{dh}{\sqrt{h}} = -\frac{A_e\sqrt{2g}}{A_T}\int_0^{t_{\text{drain}}} dt \implies \mathbf{t_{\text{drain}} = \frac{2A_T\sqrt{H_0}}{A_e\sqrt{2g}}}$$

CANONICAL PROBLEM WALKTHROUGH · CL1 SETUP

Airfoil Wake Deficit: Determining Upstream Capture Width

Physical Setup: An airfoil is tested in a uniform wind tunnel flow of velocity v_∞ and density ρ. Far downstream, boundary layer skin friction and wake mixing generate a symmetric velocity deficit over a wake band of width $2b$ ($|y| \le b$):

$$v(y) = v_\infty\left[1 - \Delta\left(1 - \frac{|y|}{b}\right)\right] \quad \text{for } |y| \le b; \qquad v(y) = v_\infty \quad \text{for } |y| > b$$

where $0 < \Delta < 1$ is the centerline fractional velocity defect. Far upstream where the flow is completely uniform ($v = v_\infty$), all the fluid that eventually exits through the wake $[-b, b]$ passed through a capture band $[-y_{-\infty}, y_{-\infty}]$.

Objective: Determine the upstream capture semi-width $y_{-\infty}$ from conservation of mass.

Stage 1 Select the Wake Streamtube Control Volume

Define a 2D control volume bounded by:

- **Upstream inlet face (Σ_{in}):** Located at $x = -\infty$, extending across $y \in [-y_{-\infty}, y_{-\infty}]$ where velocity is uniform $v_x = v_\infty$.
- **Downstream exit face (Σ_{out}):** Located at $x = +\infty$, spanning the exact wake thickness $y \in [-b, b]$ with velocity profile $v(y)$.
- **Upper and lower boundaries ($\Sigma_{\text{streamline}}$):** The limiting streamlines that pass through $(\pm b)$ at downstream infinity. By definition, *no fluid crosses a streamline*: $(\vec{v} \cdot \vec{n}) \equiv 0$.

Stage 2 Evaluate Upstream Inflow Mass Rate

For unit span ($W = 1$) into the page, integrating over the upstream aperture:

$$\dot{m}_{\text{in}} = \int_{-y_{-\infty}}^{y_{-\infty}} \rho v_\infty \, dy = 2\rho v_\infty y_{-\infty}$$

Stage 3 Evaluate Downstream Wake Outflow Mass Rate

Exploiting symmetry between upper and lower halves of the wake:

$$\dot{m}_{\text{out}} = \int_{-b}^{b} \rho v(y) \, dy = 2\rho v_\infty \int_0^b \left[1 - \Delta \left(1 - \frac{y}{b}\right)\right] dy$$

Evaluating the spatial integral across the shear layer:

$$\int_0^b \left[1 - \Delta + \Delta \frac{y}{b}\right] dy = \left[(1-\Delta)y + \Delta \frac{y^2}{2b}\right]_0^b = (1-\Delta)b + \frac{\Delta b}{2} = b\left(1 - \frac{\Delta}{2}\right)$$

Therefore, the total wake mass flow rate is:

$$\dot{m}_{\text{out}} = 2\rho v_\infty b \left(1 - \frac{\Delta}{2}\right)$$

Stage 4 Solve for Upstream Width $y_{-\infty}$

Since the flow is steady and no mass escapes through the bounding streamlines, $\dot{m}_{\text{in}} = \dot{m}_{\text{out}}$:

$$2\rho v_\infty y_{-\infty} = 2\rho v_\infty b \left(1 - \frac{\Delta}{2}\right)$$

Dividing by $2\rho v_\infty$ gives the exact analytical result:

$$\mathbf{y_{-\infty} = b\left(1 - \frac{\Delta}{2}\right)}$$

AEROSPACE EXAM WARNING: THE LATERAL LEAKAGE TRAP

Notice that because $\Delta > 0$, we have $y_{-\infty} < b$! Because fluid in the wake has been slowed down by aerodynamic drag, each fluid package moves slower, requiring a larger cross-sectional area to carry the same mass flux. Consequently, **streamlines diverge around and behind the airfoil**.

If you choose a rigid rectangular control volume of fixed height $2H > 2b$ instead of streamline boundaries, fluid *must leak outward through the top and bottom surfaces* with mass rate $\dot{m}_{\text{lateral}} = 2\rho v_\infty (b - y_{-\infty}) = \rho v_\infty b\Delta$. Failing to include this lateral mass escape is the most frequent exam mistake in wake momentum balances!